4. Consider the following code snippet:

```
int main(int argc, char *argv[])
{
    void *my_mem = 0;
    int alloc_blocks = 0;

    while(1)
    {
        my_mem = (void *) malloc(1024 * 1024);
        if (!my_mem) break;
        memset(my_mem,1, (1024*1024));
        printf("allocated memory blocks is %d MB\n",++alloc_
blocks);
    }
    exit(0);
}
```

Can you explain what will happen when you compile and run the program?

5. In the same snippet, assume that you are asked to modify the code so as to remove the call to the memset function. Can you explain what will happen when you compile and run the code?
6. You are given four cards, each with a digit on one side and an alphabet on the other. The hypothesis you need to validate is that if a card has a vowel on one side, then it has an even number on the other side. The four cards are laid out such that Card 1 shows A, Card 2 shows D, Card 3 shows 4 and Card 4 shows 7. You need to specify which of these four cards are needed to check whether this rule is true or not.
7. You have been given all the stock market transactions made by a user for buying and selling shares. Each transaction includes the following details: (a) name of the company, (b) type of transaction — buying/selling, (c) number of shares bought/sold, (d) price of each share bought/sold, and (e) date of the transaction. You also have access to the stock market information in terms of the price of each company's stock at the end of every day. A user would like to know which of his shares he can sell at a profit of 10 per cent or above on each day. You are asked to write a program that examines the closing price of the previous day's stock market prices, and provide recommendations to the user on which of his equity holdings he can sell that day for a profit of 10 per cent or more. Your recommendation will contain the company's name, number of shares to sell, the sale price and the percentage of profit that the user will make if he sells at that price. What is the time complexity of your algorithm, assuming that you are given N transactions in total, corresponding to M different companies?
8. Consider the above question. Stock markets are quite volatile and the price of company stocks can fluctuate considerably during the day. You are now told that instead of using the previous day's closing price, your application should read in real-time the current price of the stocks that the user holds, and then recommend which of them he can sell at that point of time for a 10 per cent profit. Assume that you have stock market prices available as a real-time feed every 60 seconds. How will you modify your solution for Question 7 to satisfy this criterion?
9. You are given an array of N records containing employee information, including the date of joining, name, address and salary. You are told to sort the array in ascending order (in relation to the employee's date of joining). It is possible that multiple employee records may have the same joining date. In that case, the original order of these records should be retained in the final sorted array. (a) If you are told that there is no restriction on the additional space that your algorithm can use, what will your solution be? (b) And if you are told that you can only use constant additional storage space, what will your solution be then?
10. You are given a search query and can use any of your favourite Web search engines to find the query results. Web search engines use their own algorithm to rank the query search results. However, you are asked to apply a filter on top of the search engine query results such that the results are ordered temporally. For example, consider an example query of the form: 'Donald Trump US Presidential election'. Given a set of search engine results for this query which will contain news articles, events, etc, you need to sort them temporally. For instance, a query result link on the news story from July 2015 describing Trump's intention to run should be ranked before the news story featuring his 2016 New Year wishes to the American electorate. Essentially, your task is to organise the search engine query results on a timeline. What are the challenges in organising the Web search engine query results temporally?
11. You are given an array of N integers. You are then asked to find the largest element that appears an even number of times in the array. What is the time complexity of your algorithm? Can you do this without sorting the entire array?
12. Given a stream of input integers in real-time, you are asked to find the minimum element in the stream at any point in time. What is the complexity of your algorithm? How will your solution change (a) if you are asked to find both minimum and maximum elements in the stream at any point in time, and (b) if you are asked to find both the 'first minimum' and 'second minimum' in the stream at any point in time?
13. Given a singly linked list, you are asked to write a function to find the middle element of the linked list. You are not given the length of the list. How will you modify your solution if the given linked list is a singly linked circular list?

Feel free to reach out to me over LinkedIn/email if you need any help in your new study journey. If you have any favourite programming questions/software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at *sandyasm_AT_yahoo_DOT_com*. Wishing all our readers happy coding until next month! END

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